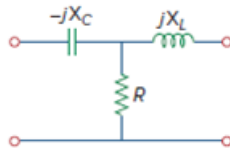


Problem

Using Fig. 19.97, design a problem to help other students better understand how to find g parameters in an ac circuit.

Figure 19.97



Step-by-step solution

Step 1 of 7

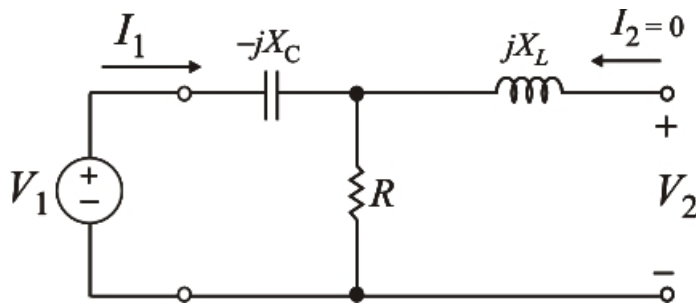
The g parameters are

$$I_1 = g_{11}V_1 + g_{12}I_2$$

$$V_2 = g_{21}V_1 + g_{22}I_2$$

To obtain g_{11} and g_{21} , we open-circuit the output port and connect a voltage source up to the input port.

Step 2 of 7



Step 3 of 7

Input current is,

$$I_1 = \frac{V_1}{R - jX_c}$$

$$\frac{I_1}{V_1} = \frac{1}{R - jX_c}$$

We know that,

$$g_{11} = \left. \frac{I_1}{V_1} \right|_{I_2=0}$$

$$\therefore g_{11} = \frac{1}{R - jX_c}$$

Step 4 of 7

By voltage division, we get

$$V_2 = \left(\frac{R}{R - jX_C} \right) V_1$$

$$\frac{V_2}{V_1} = \frac{R}{R - jX_C}$$

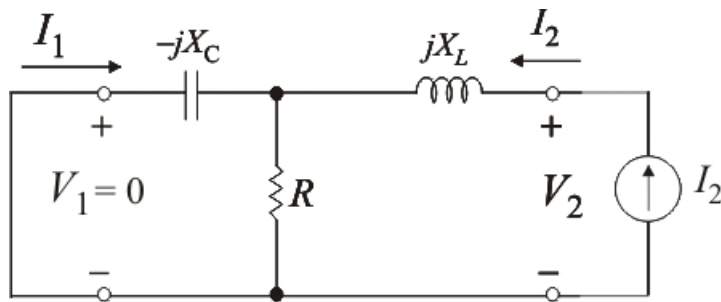
We know that,

$$g_{21} = \left. \frac{V_2}{V_1} \right|_{I_2=0}$$

$$\therefore g_{21} = \frac{R}{R - jX_C}$$

Step 5 of 7

To get g_{12} and g_{22} , we short – circuit input port and connect a current source I_2 to the output port



Step 6 of 7

By current division, we obtain

$$I_1 = \left(\frac{-R}{R - jX_C} \right) I_2$$

$$\frac{I_1}{I_2} = \left(\frac{-R}{R - jX_C} \right)$$

$$\frac{I_1}{I_2} = \frac{-R}{R - jX_C}$$

We know that,

$$g_{12} = \left. \frac{I_1}{I_2} \right|_{V_1=0}$$

$$\therefore g_{12} = \frac{-R}{R - jX_C}$$

Step 7 of 7

Where $R_1 - jX_C$ are in parallel and in series with jX_L , then

$$\Rightarrow \frac{(R)(-jX_C)}{R - jX_C}$$

$$\Rightarrow \frac{-jRX_C}{R - jX_C} + jX_L$$

The output voltage is

$$V_2 = \left[\frac{-jRX_C}{R - jX_C} + jX_L \right] I_2$$

$$\frac{V_2}{I_2} = \frac{-jRX_C}{R - jX_C} + jX_L$$

We know that,

$$g_{22} = \left. \frac{V_2}{I_2} \right|_{V_1=0}$$

$$\therefore g_{22} = \frac{jRX_C}{R - jX_C} + jX_L$$

Thus, the g parameters in matrix form is

$$[g] = \begin{bmatrix} \frac{1}{R - jX_C} & \frac{-R}{R - jX_C} \\ \frac{R}{R - jX_C} & \frac{-jRX_C}{R - jX_C} + jX_L \end{bmatrix}$$